

Computer Organisation

Topic 13

Multiprocessor Organisation

Outline

- Multiple Processor Organizations
- Symmetric Multiprocessors
- Cache Coherence and the MESI Protocol
- Clusters
- Memory Access
- Multicore Organizations

Multiple Processor Organization

- Flynn's Classification
 - Single instruction, single data stream - SISD
 - Single instruction, multiple data stream - SIMD
 - Multiple instruction, single data stream - MISD
 - Multiple instruction, multiple data stream - MIMD

Single Instruction, Single Data Stream - SISD

- Single processor
- Single instruction stream
- Data stored in single memory
- Uni-processor

Single Instruction, Multiple Data Stream - SIMD

- Single machine instruction
- Controls simultaneous execution
- Number of processing elements
- Lockstep basis
- Each processing element has associated data memory
- Each instruction executed on different set of data by different processors
- Vector and array processors

Multiple Instruction, Single Data Stream - MISD

- Sequence of data
- Transmitted to set of processors
- Each processor executes different instruction sequence
- Never been implemented

Multiple Instruction, Multiple Data Stream- MIMD

- Set of processors
- Simultaneously execute different instruction sequences
- Different sets of data
- SMPs, clusters and NUMA systems

MIMD - Overview

- General purpose processors
- Each can process all instructions necessary
- Further classified by method of processor communication

Tightly Coupled - SMP

- Processors share memory
- Communicate via that shared memory
- Symmetric Multiprocessor (SMP)
 - Share single memory or pool
 - Shared bus to access memory
 - Memory access time to given area of memory is approximately the same for each processor

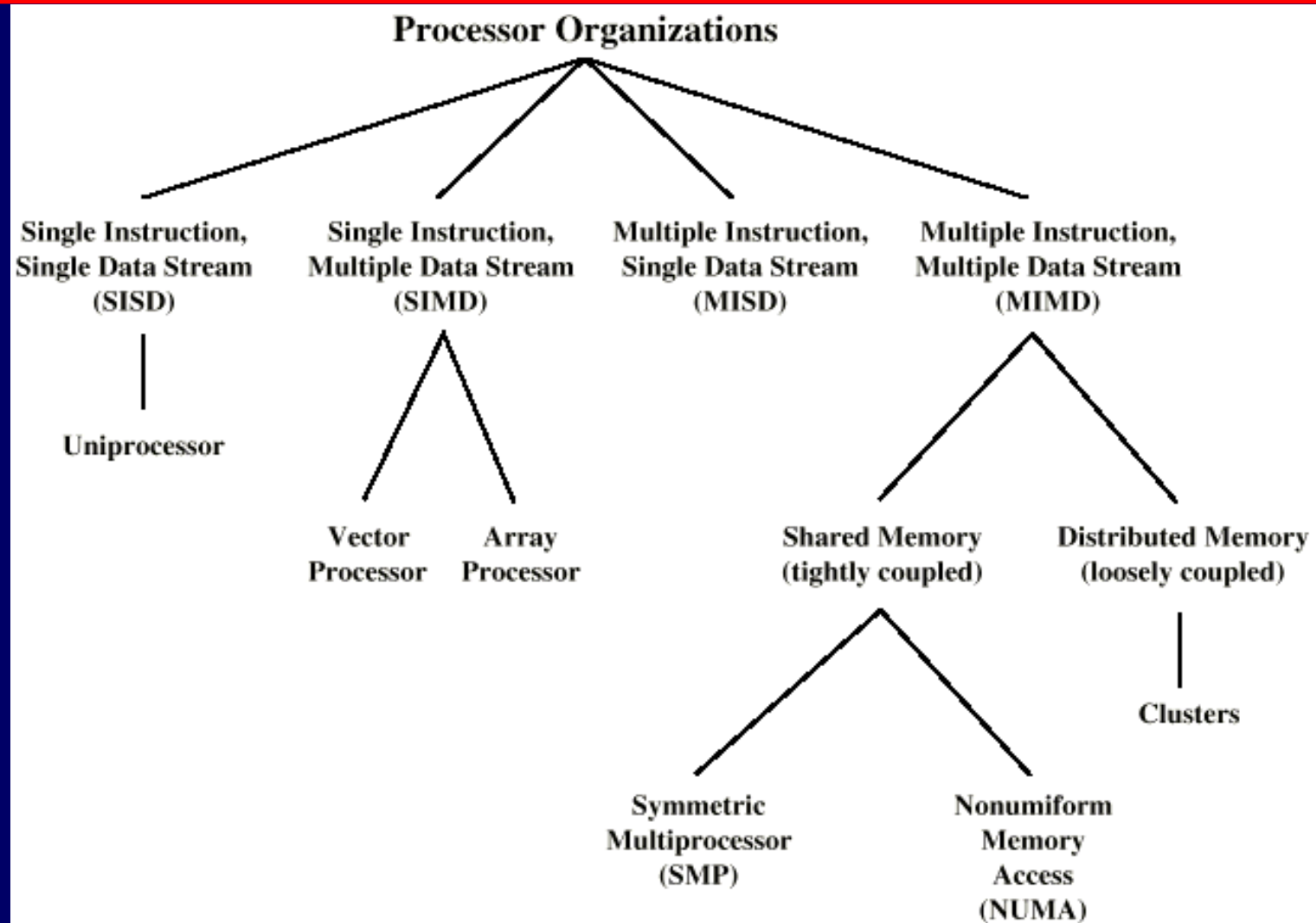
Tightly Coupled - NUMA

- Nonuniform memory access
- Access times to different regions of memory may differ

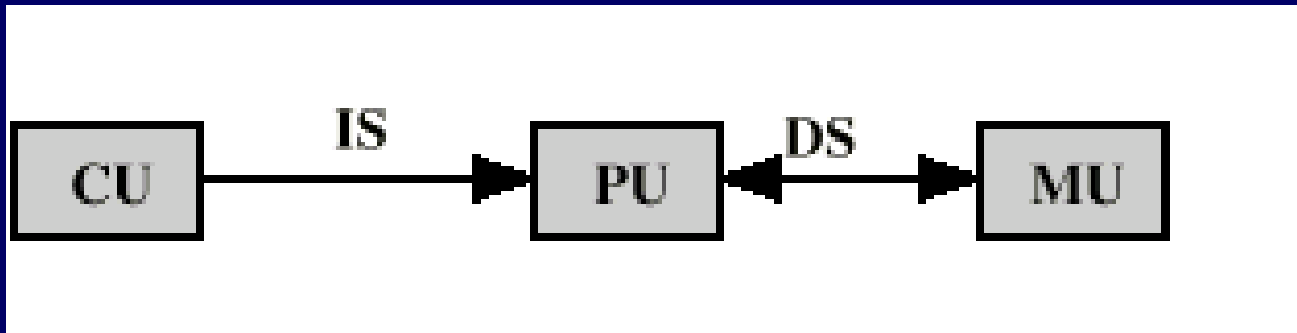
Loosely Coupled - Clusters

- Collection of independent uniprocessors or SMPs
- Interconnected to form a cluster
- Communication via fixed path or network connections

Taxonomy of Parallel Processor Architectures



Parallel Organizations - SISD



CU: Control Unit

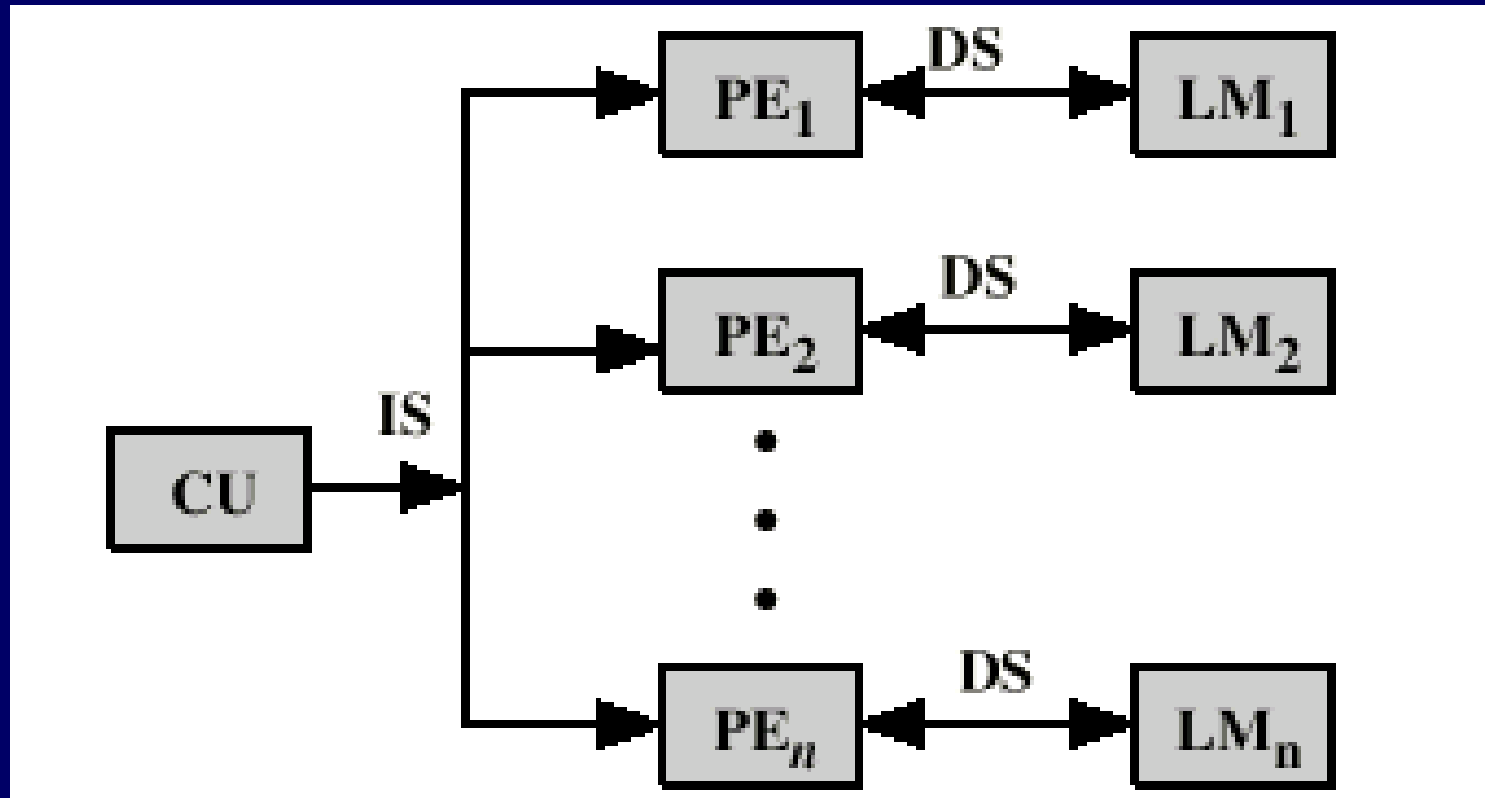
IS: Instruction Stream

PU: Processing Unit

DS: Data Stream

MU: Memory Unit

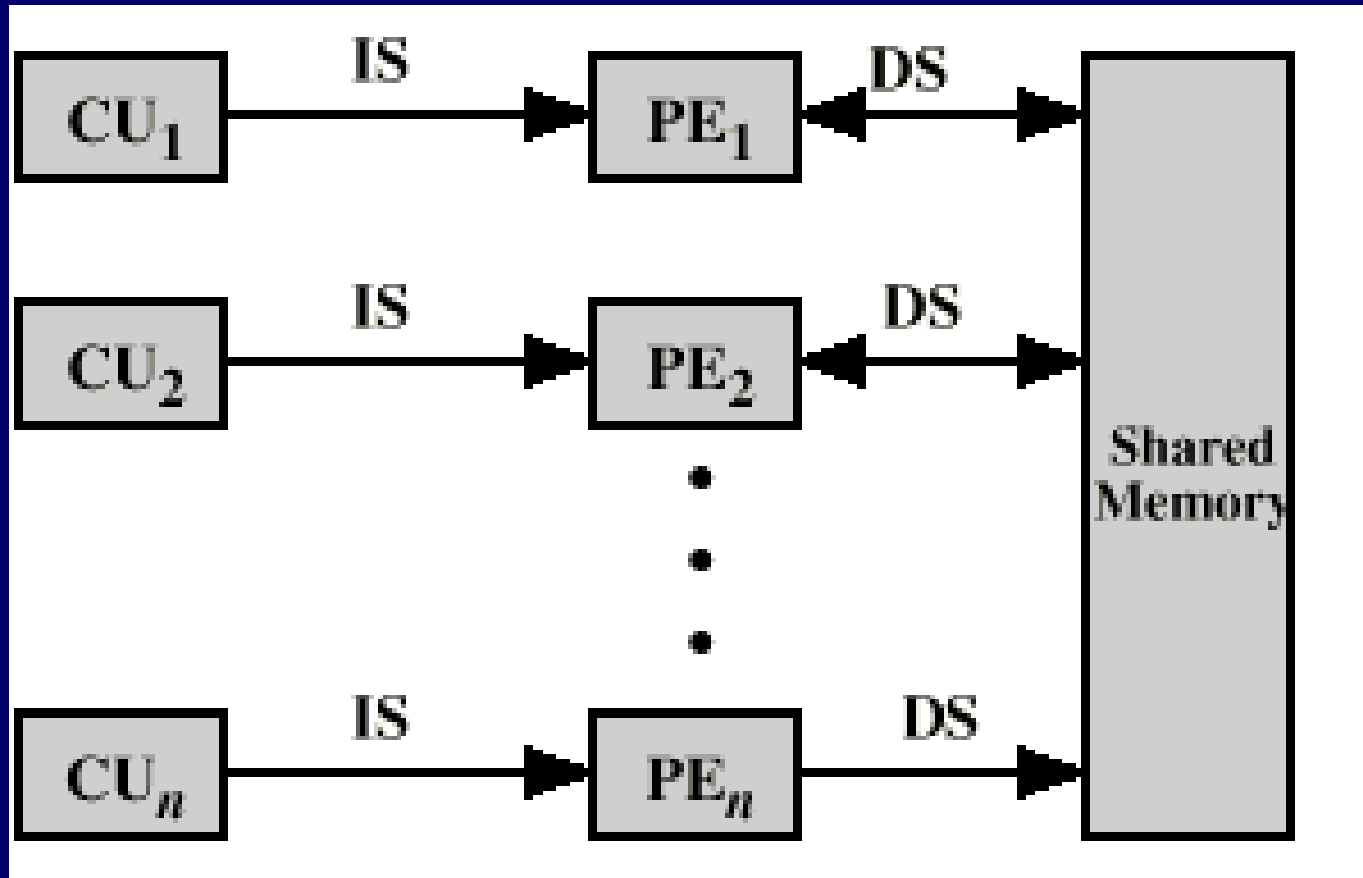
Parallel Organizations - SIMD



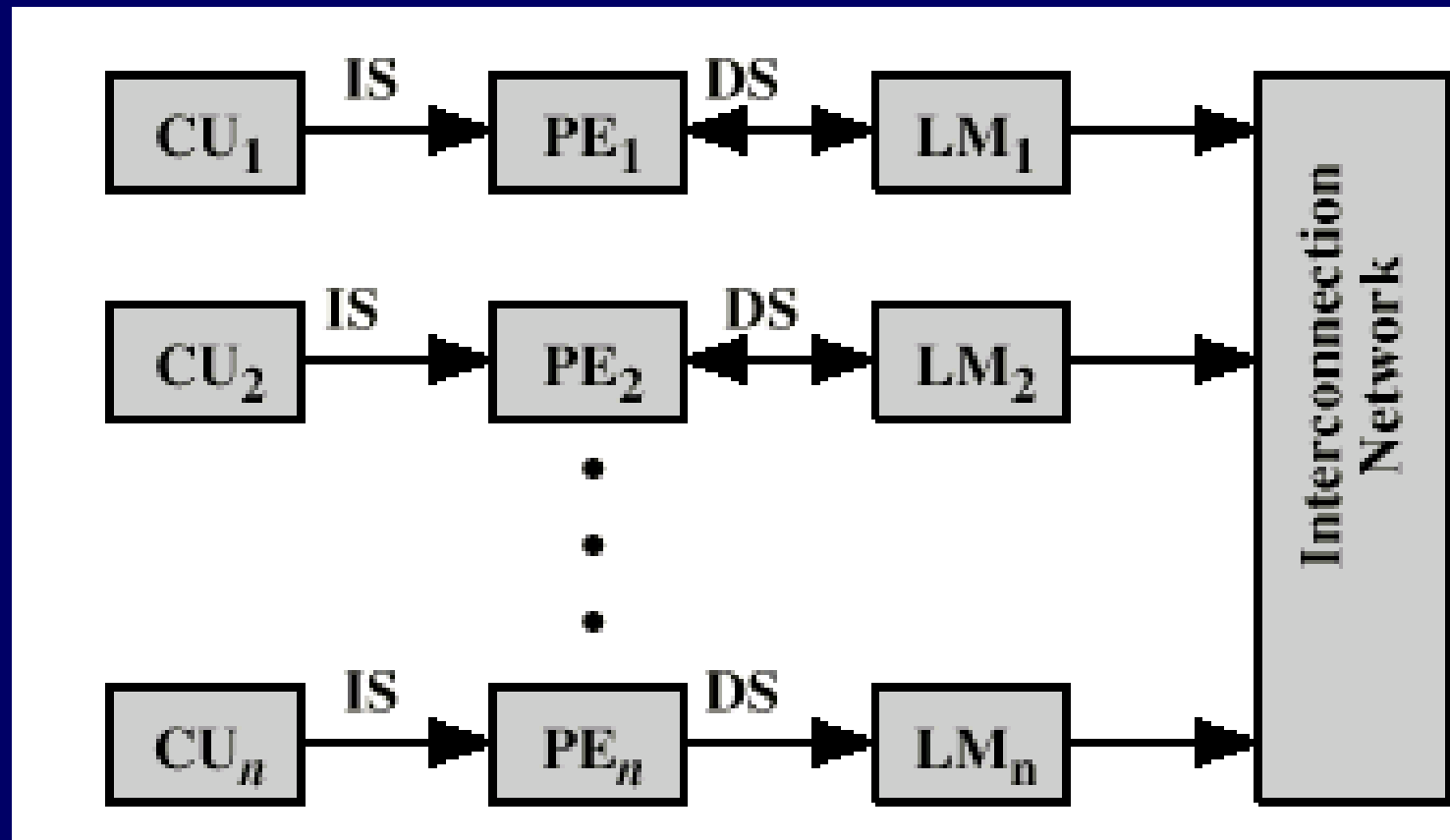
PE: Processing Element

LM: Local Memory

Parallel Organizations - MIMD Shared Memory



Parallel Organizations - MIMD Distributed Memory



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Symmetric Multiprocessors (1)

- A stand alone computer with the following characteristics
 - Two or more similar processors of comparable capacity
 - Processors share same memory and I/O
 - Processors are connected by a bus or other internal connection
 - Memory access time is approximately the same for each processor
 - All processors share access to I/O
 - Either through same channels or different channels giving paths to same devices

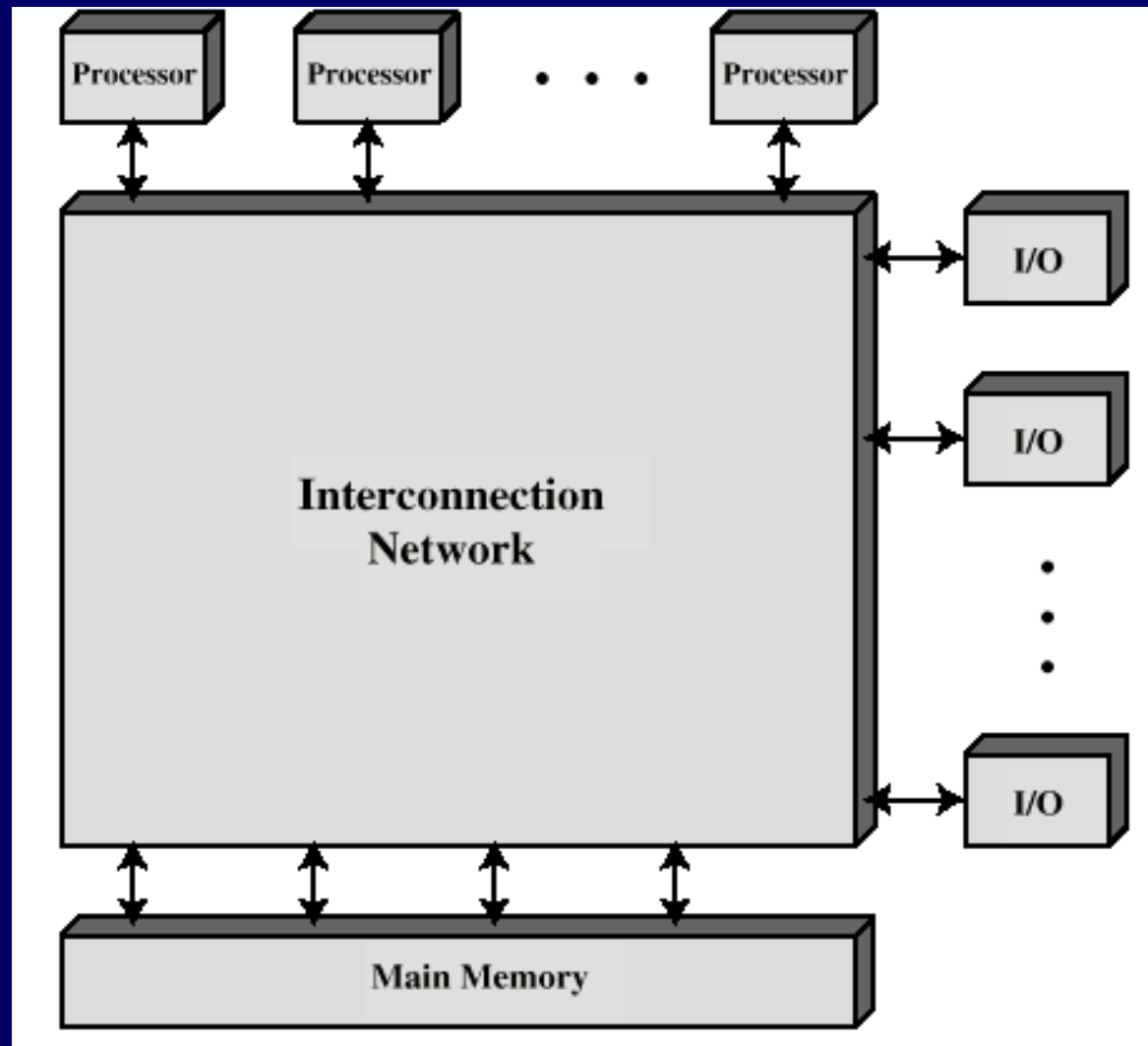
Symmetric Multiprocessors (2)

- A stand alone computer with the following characteristics (cont)
 - All processors can perform the same functions (hence symmetric)
 - System controlled by integrated operating system
 - providing interaction between processors
 - Interaction at job, task, file and data element levels

SMP Advantages

- Performance
 - If some work can be done in parallel
- Availability
 - Since all processors can perform the same functions, failure of a single processor does not halt the system
- Incremental growth
 - User can enhance performance by adding additional processors
- Scaling
 - Vendors can offer range of products based on number of processors

Organisation: Block Diagram of Tightly Coupled Multiprocessor



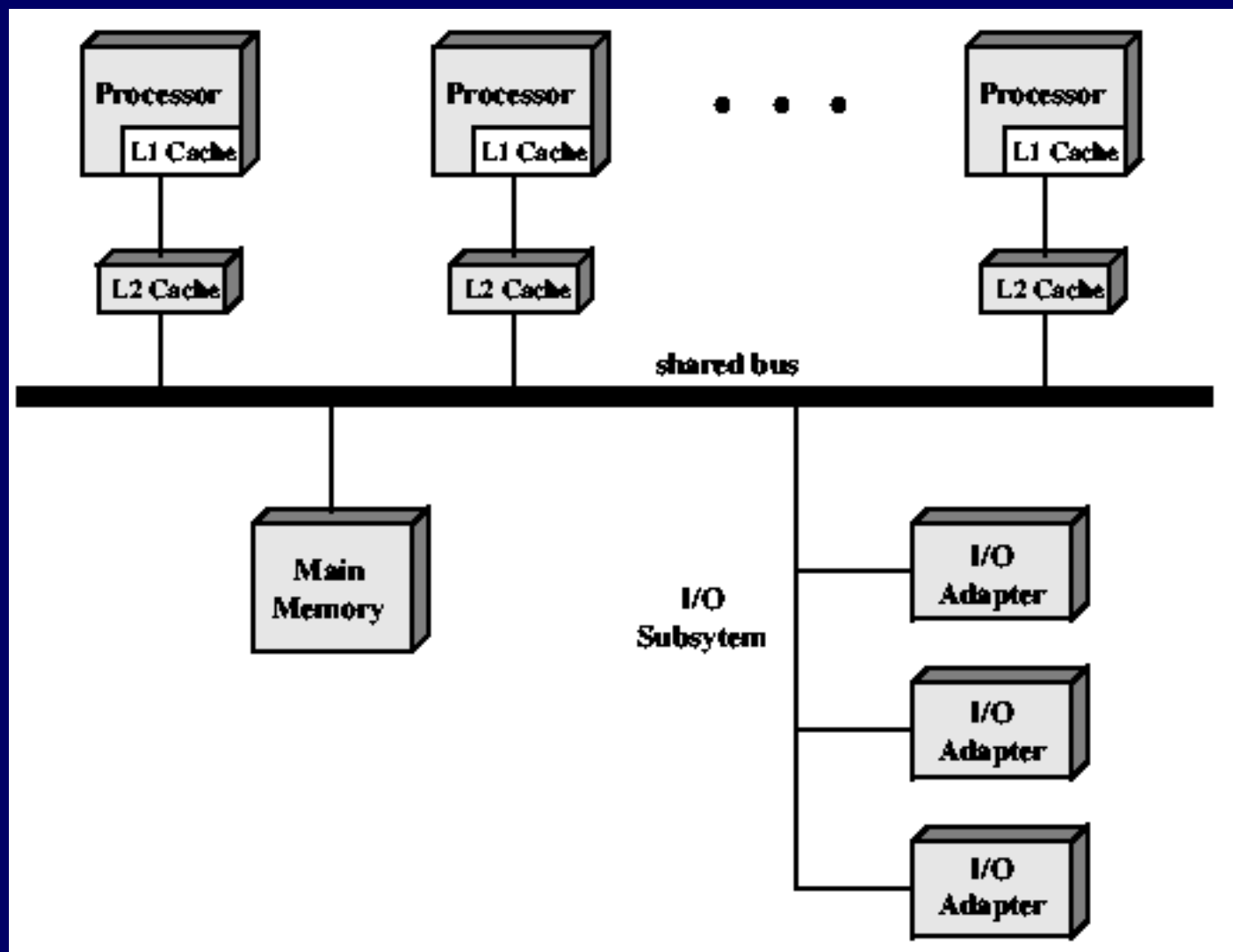
Organization Classification

- Time shared or common bus
- Multiport memory
- Central control unit

Time Shared Bus

- Simplest form
- Structure and interface similar to single processor system
- Following features provided
 - Addressing - distinguish modules on bus
 - Arbitration - any module can be temporary master
 - Time sharing - if one module has the bus, others must wait and may have to suspend
- Now have multiple processors as well as multiple I/O modules

Shared Bus



Time Shared Bus - Advantages

- Simplicity
- Flexibility
- Reliability

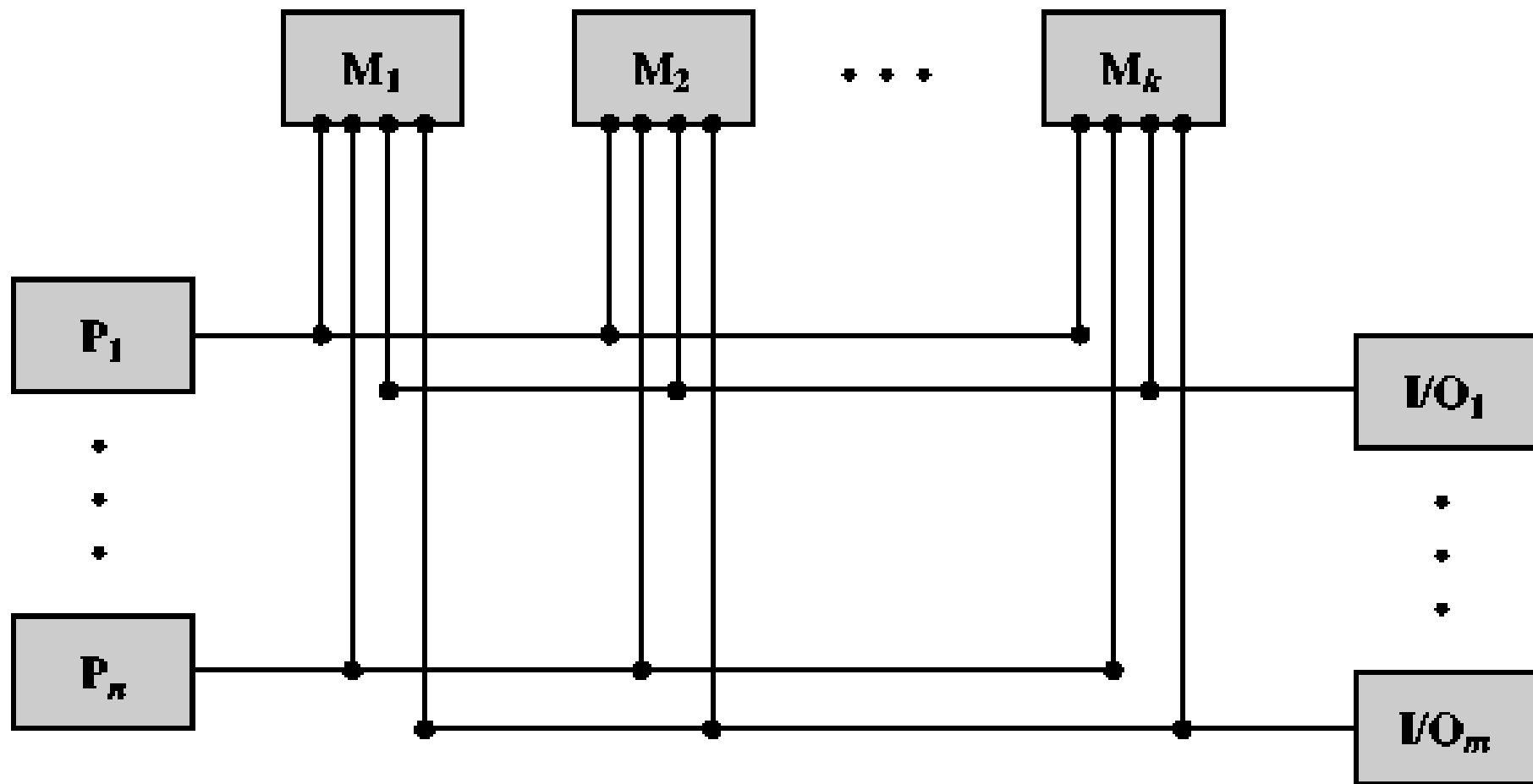
Time Share Bus - Disadvantage

- Performance limited by bus cycle time
- Each processor should have local cache
 - Reduce number of bus accesses
- Leads to problems with cache coherence
 - Solved in hardware - see later

Multiport Memory

- Direct independent access of memory modules by each processor
- Logic required to resolve conflicts
- Little or no modification to processors or modules required

Multiport Memory Diagram



Multiport Memory - Advantages and Disadvantages

- More complex
 - Extra logic in memory system
- Better performance
 - Each processor has dedicated path to each module
- Can configure portions of memory as private to one or more processors
 - Increased security
- Write through cache policy

Central Control Unit

- Funnels separate data streams between independent modules
- Can buffer requests
- Performs arbitration and timing
- Pass status and control
- Perform cache update alerting
- Interfaces to modules remain the same
- e.g. IBM S/370

Operating System Issues

- Simultaneous concurrent processes
- Scheduling
- Synchronization
- Memory management
- Reliability and fault tolerance

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Cache Coherence and MESI Protocol

- Problem - multiple copies of same data in different caches
- Can result in an inconsistent view of memory
- Write back policy can lead to inconsistency
- Write through can also give problems unless caches monitor memory traffic

Software Solutions

- Compiler and operating system deal with problem
- Overhead transferred to compile time
- Design complexity transferred from hardware to software
- However, software tends to make conservative decisions
 - Inefficient cache utilization
- Analyze code to determine safe periods for caching shared variables

Hardware Solution

- Cache coherence protocols
- Dynamic recognition of potential problems
- Run time
- More efficient use of cache
- Transparent to programmer
- 2 types of hardware protocols:
 - Directory protocols
 - Snoopy protocols

Directory Protocols

- Collect and maintain information about copies of data in cache
- Directory stored in main memory
- Requests are checked against directory
- Appropriate transfers are performed
- Creates central bottleneck
- Effective in large scale systems with complex interconnection schemes/multiple buses

Snoopy Protocols

- Distribute cache coherence responsibility among cache controllers
- Cache recognizes that a line is shared
- Updates announced to other caches
 - Each cache controller 'snoops' on the network
- Suited to bus based multiprocessor
 - Shared bus used for broadcasting and snooping
- Increases bus traffic
- 2 approaches:
 - Write invalidate
 - Write update

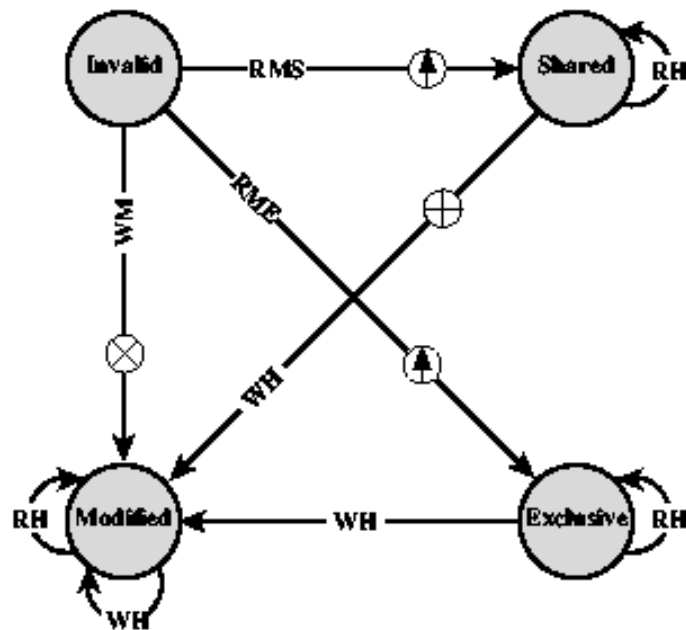
Write Invalidate

- Multiple readers, one writer
- When a write is required, all other caches of the line are invalidated
- Writing processor then has exclusive (cheap) access until line required by another processor
- Used in Pentium II and PowerPC systems
- State of every line is marked as modified, exclusive, shared or invalid
- MESI

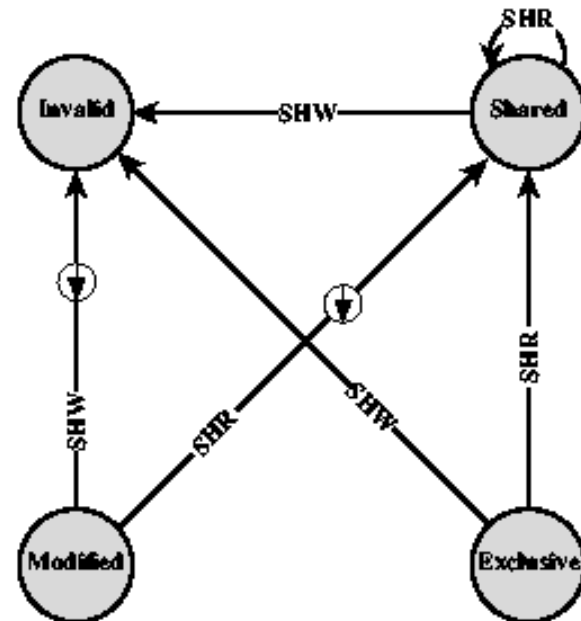
Write Update

- Multiple readers and writers
- Updated word is distributed to all other processors
- Some systems use an adaptive mixture of both solutions

MESI State Transition Diagram



(a) Line in cache at initiating processor



(b) Line in snooping cache

RH Read hit
RMS Read miss, shared
RME Read miss, exclusive
WH Write hit
WM Write miss
SHR Snoop hit on read
SHW Snoop hit on write or read-with-intent-to-modify

Dirty line copyback
 Invalidate transaction
 Read-with-intent-to-modify
 Cache line fill

MESI Protocol Example: Read Miss (Invalid)

- If **one** other cache has clean copy of line in **exclusive** state, it changes from **exclusive** to **shared**
 - And initiating processor changes cache from **invalid** to **shared**
- If one or **more** cache has clean copy of line in **shared** state
 - Initiating processor changes cache from **invalid** to **shared**
- If **one** other cache has **modified** the line, it blocks memory read and changes from **modified** to **shared**
- If **no** other cache has copy, processor reads line and changes from **invalid** to **exclusive**

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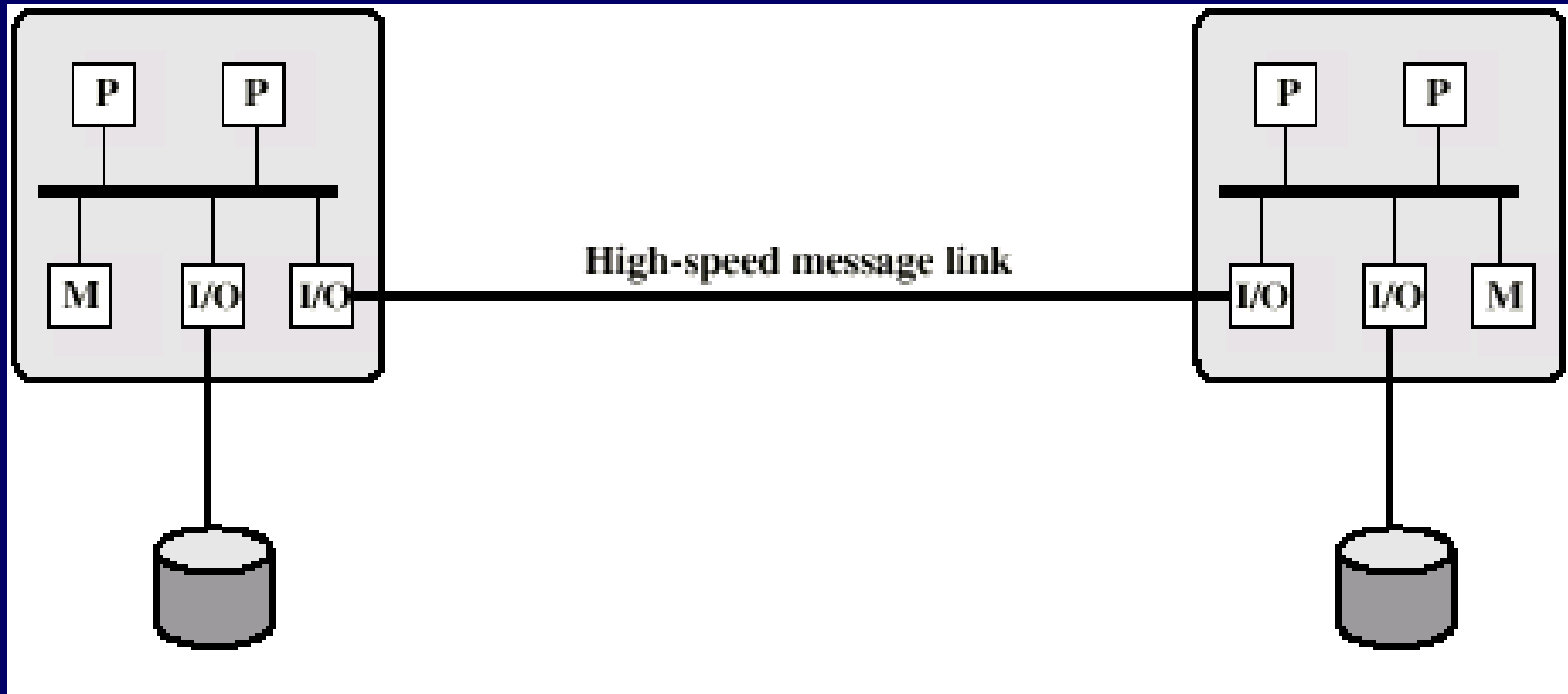
Clusters

- Alternative to SMP
 - High performance
 - High availability
 - Server applications
-
- A group of interconnected whole computers
 - Working together as unified resource
 - Illusion of being one machine
 - Each computer called a node

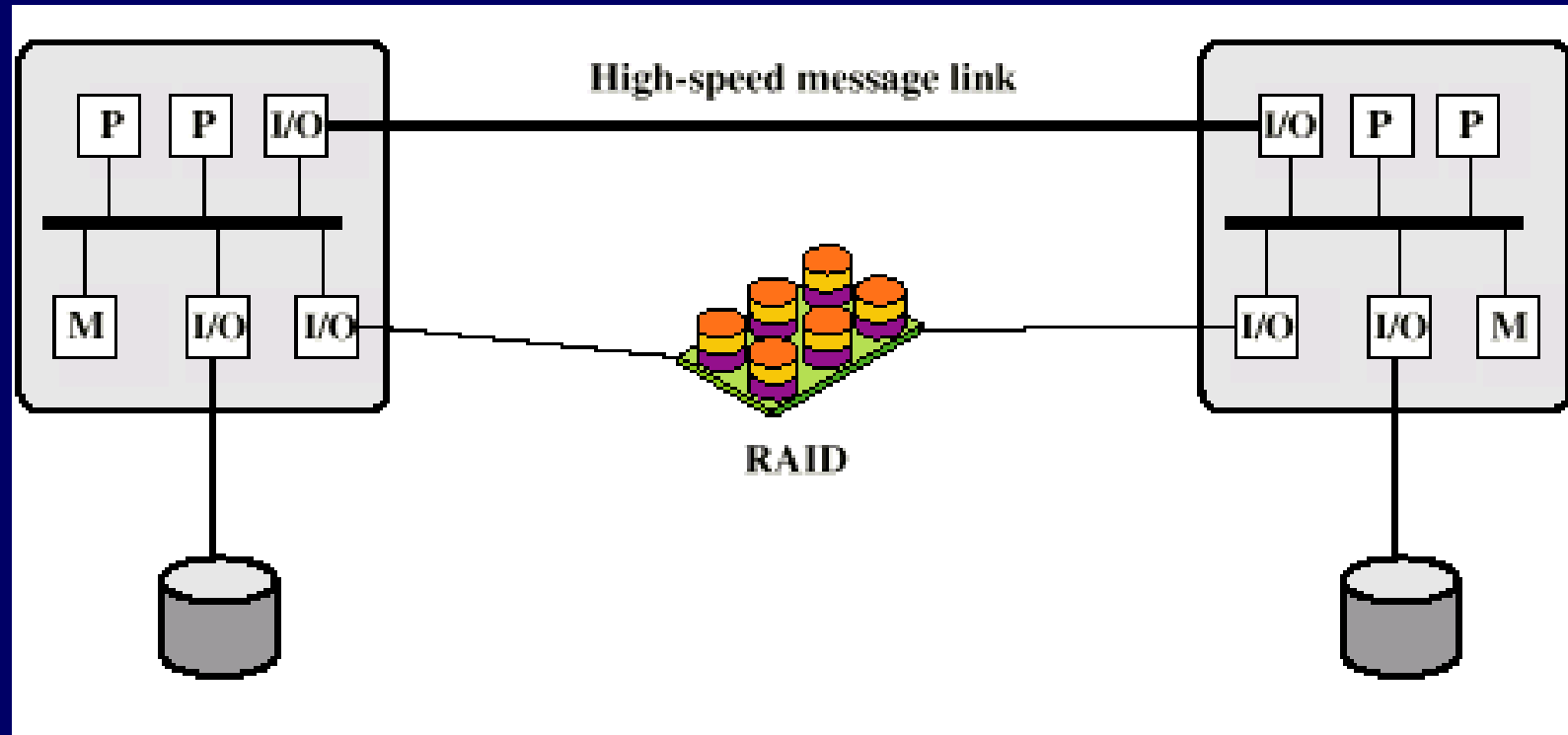
Cluster Benefits

- Absolute scalability
- Incremental scalability
- High availability
- Superior price/performance

Cluster Configurations - Standby Server, No Shared Disk



Cluster Configurations - Shared Disk



Operating Systems Design Issues (1)

- Failure Management
 - High availability
 - Fault tolerant
 - Failover
 - Switching applications & data from failed system to alternative within cluster
 - Failback
 - Restoration of applications and data to original system
 - After problem is fixed

Operating Systems Design Issues (2)

- Load balancing
 - Incremental scalability
 - Automatically include new computers in scheduling
 - Middleware needs to recognise that processes may switch between machines

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Uniform Memory Access

- All processors have access to all parts of memory
 - Using load & store
- Access time to all regions of memory is the same
- Access time to memory for different processors same
- As used by SMP

Nonuniform Memory Access (NUMA)

- All processors have access to all parts of memory
 - Using load & store
- Access time of processor differs depending on region of memory
- Different processors access different regions of memory at different speeds

Outline

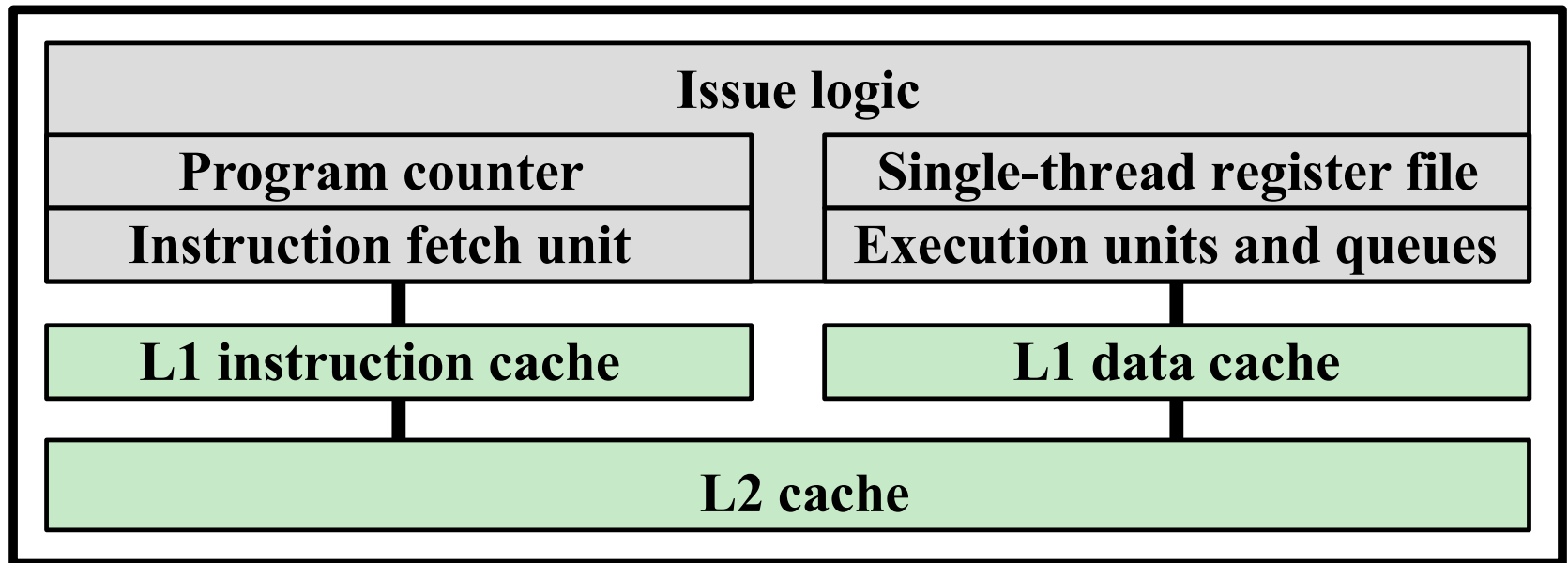
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Multicore Computer

- Core
 - “Brain” of the CPU
 - Receives instructions, performs calculations or operations, etc.
- A CPU may have multiple cores.
 - 2 cores = dual-core
 - 4 cores = quad-core
 - etc.

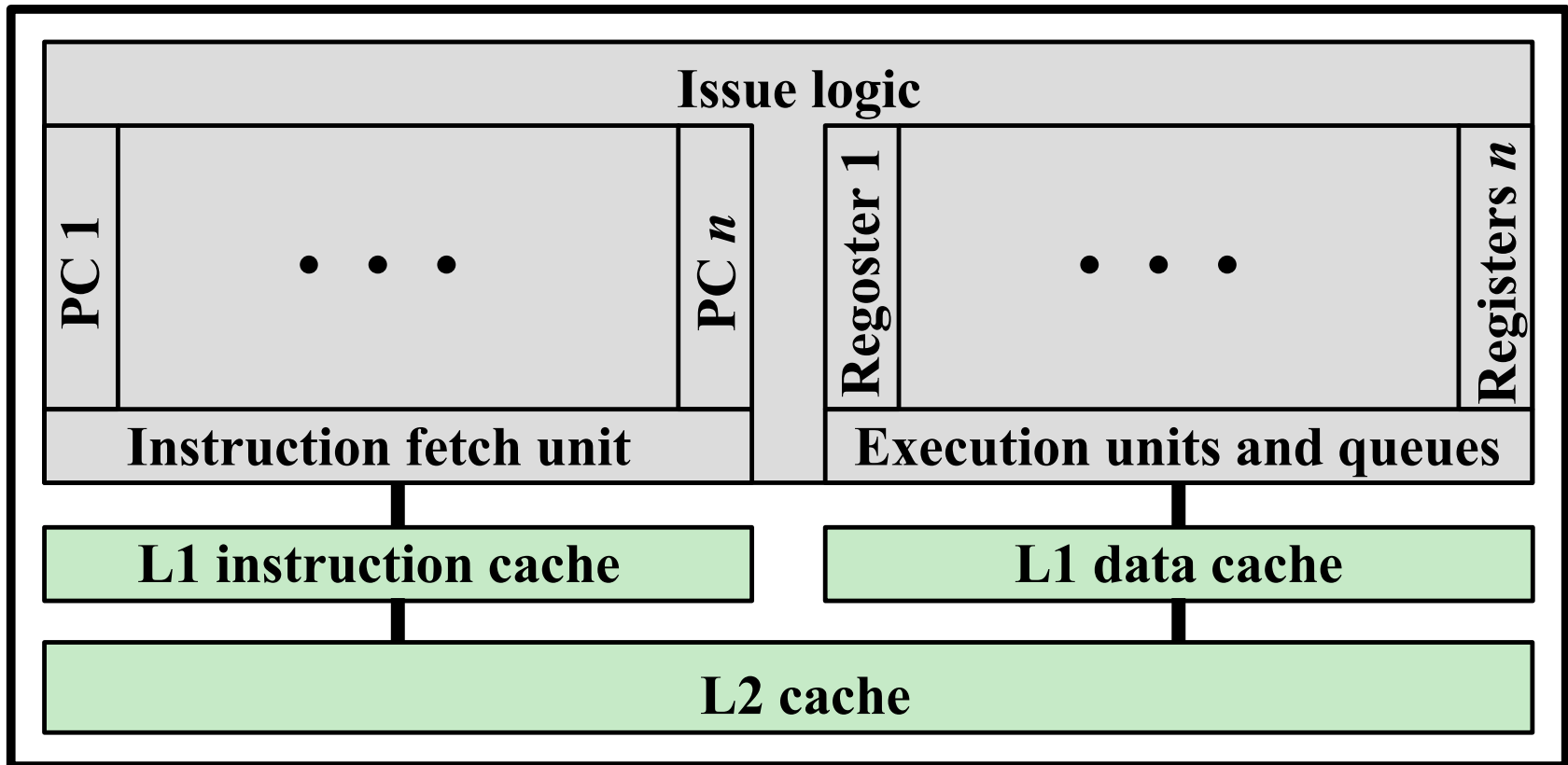
Alternative Chip Organizations

- Superscalar



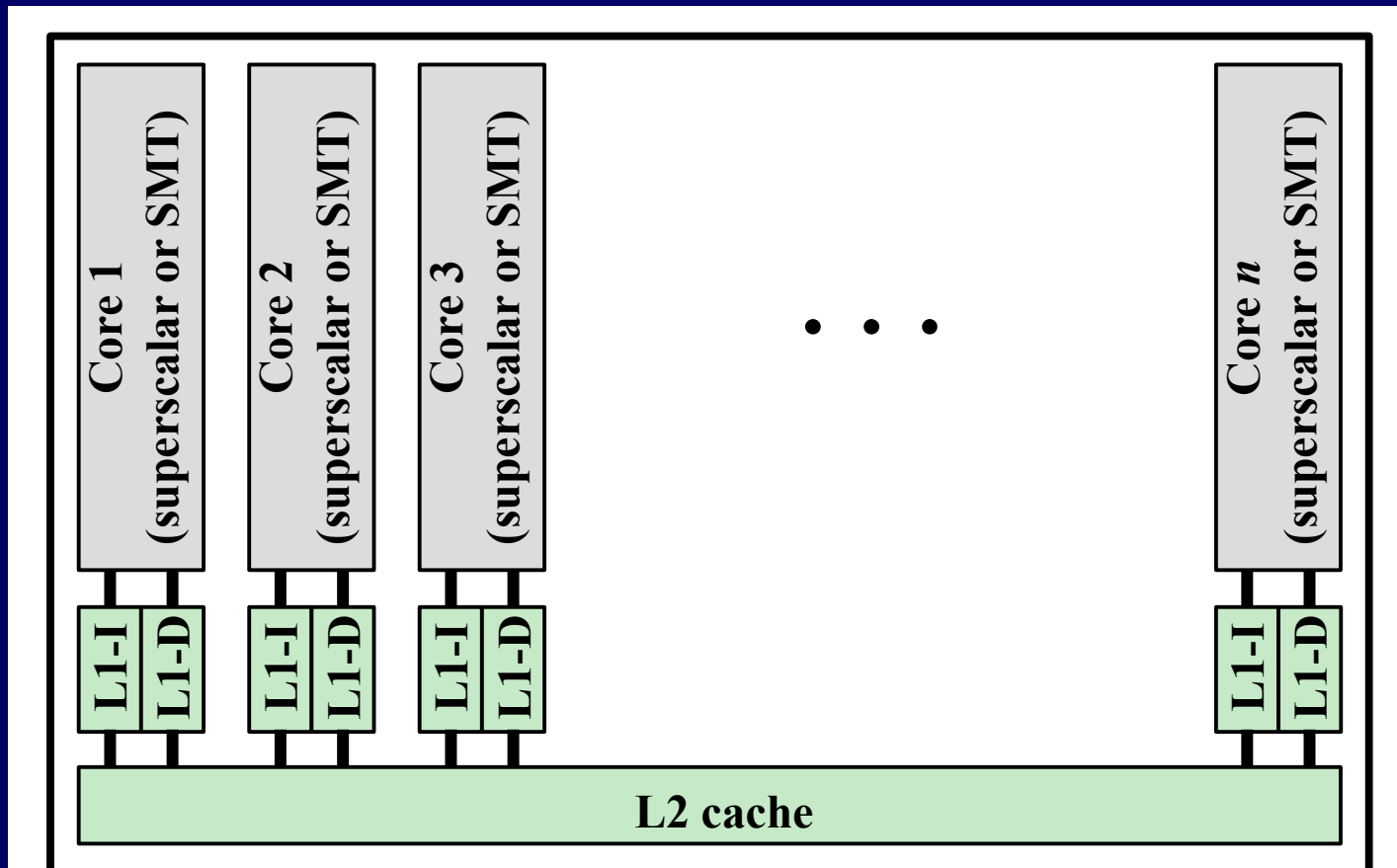
Alternative Chip Organizations

- Simultaneous multithreading

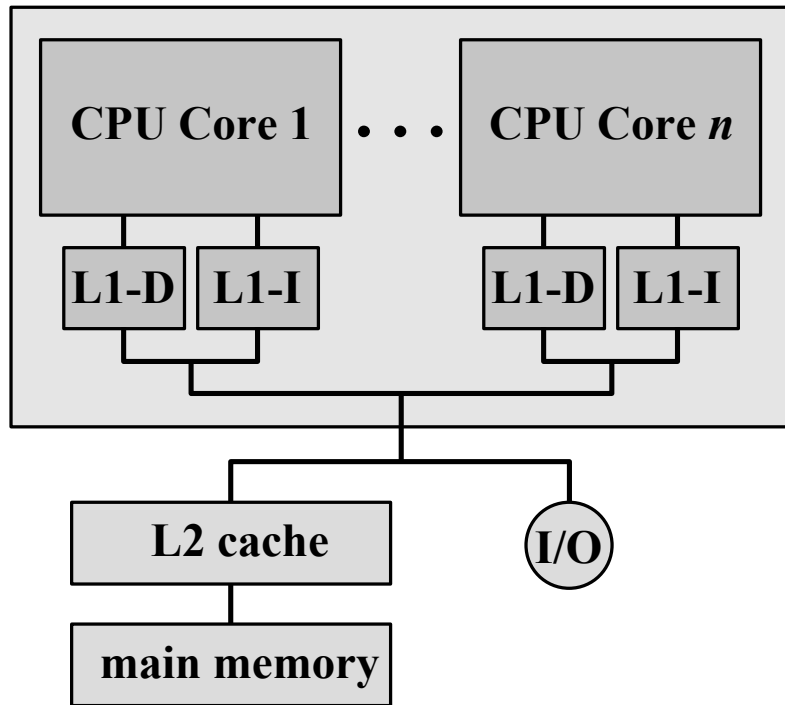


Alternative Chip Organizations

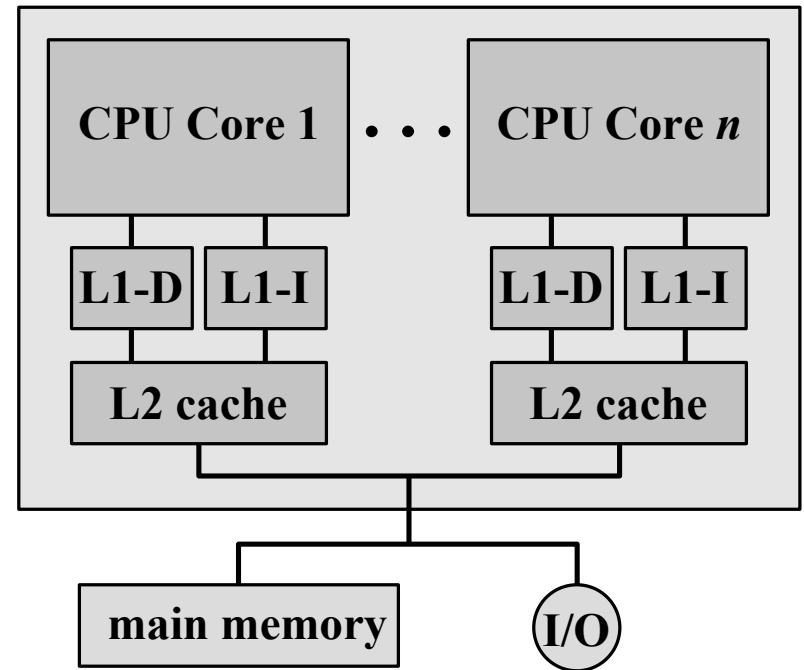
- Multicore



Multicore Organization Alternatives

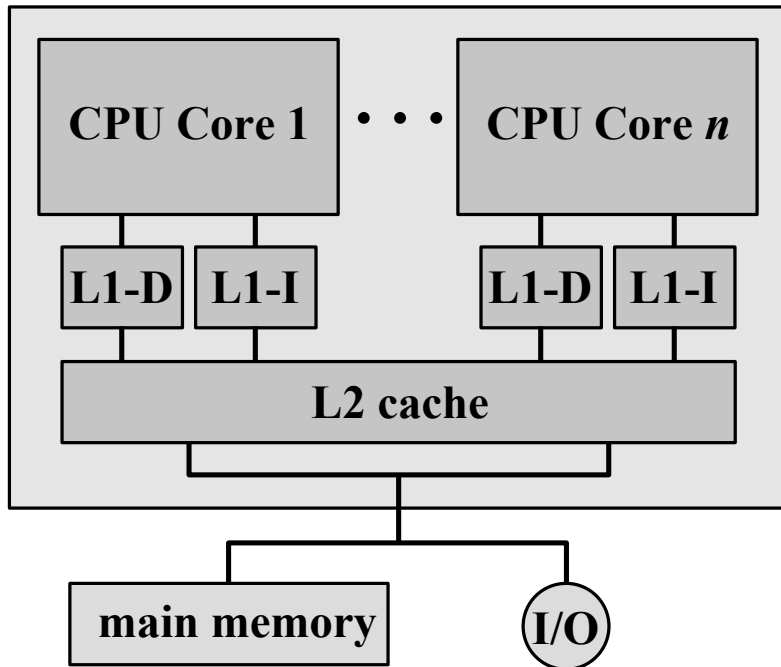


(a) Dedicated L1 cache

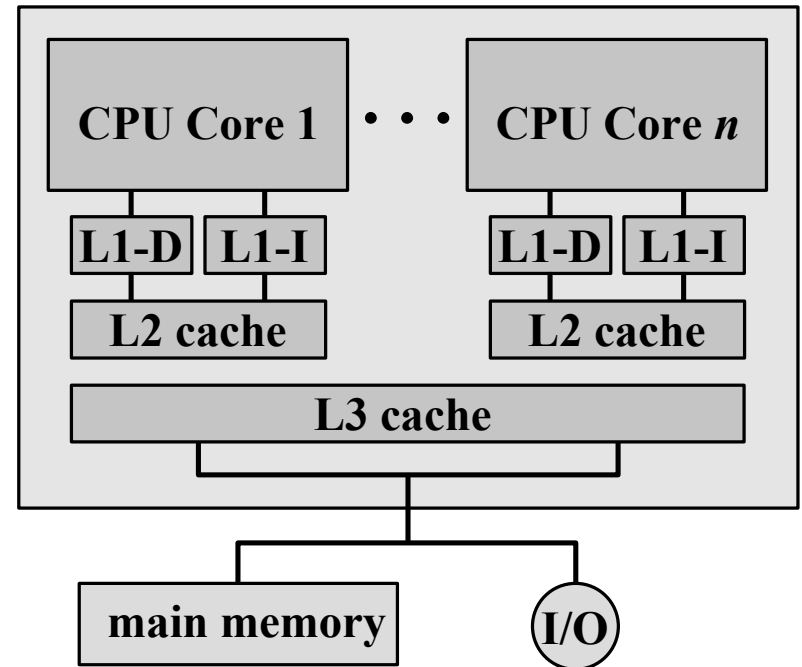


(b) Dedicated L2 cache

Multicore Organization Alternatives



(c) Shared L2 cache



(d) Shared L3 cache

Issues

- Cache coherence!

Multicore Applications

- Database servers
- Web servers
- Compilers
- Multimedia applications
- Scientific applications (CAD/CAM)